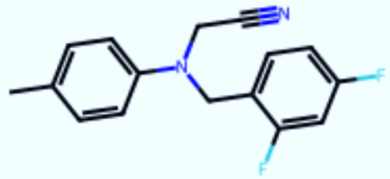
DSA103 Advanced Chemical Data Science

Lecture 13: Life Science Data – Data Formats, Tools

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	Smiles	Standard Type	Standard Relation	Standard Value	Standard Units	Assay ChEMBL ID	ROMol
...					g.mL-1	CHEMBL3625335	
	@H]1CC[C@@H]2CN1C(=O)N2OS(=O)(=O)[O-].[Na+]	MIC	'<='	0.120	ug.mL-1	CHEMBL5247824	
							

Content overview

Review SMARTS

Motivation: Data analytical tasks and challenges

Some more data formats

Useful tools:
OpenBabel
Rdkit
Biopandas
Biopython
Pymatgen

Activity: RDKit Workshop (including some recap on EDA, and a bit of Scikit-learn)

Acitivity: Data format hunt

1) What kind of data formats (with respect to your domain) come to your mind?

2) What task is the file format suited for?

3) What are some advantages and disadvantages of each?

4) If there are multiple file formats for a similar job, how would you approach conversion?

Acitivity: Excel, a classic.



The scenario: A qPCR experiment on HEK293 cells with a control group and two drugs at different doses, 15 samples in total, measured for four genes (SEPT2, MARCH1, GAPDH, ACTB).

1) Have a look at the excel file qpcr-exp14.xlsx.

2) Discuss what issues you find and how you could do that better.

3) Do that better, and do it in Pandas!

Data analytical tasks & challenges

Broad variety of tasks, e.g.:

- Prediction: Properties (QSPR), activity (QSAR), reaction outcome / yield …
- Virtual screening to find bioactive molecules (DD)
- Docking studies (protein-ligand interaction)
- Visualisation: Clustering, dimensionality reduction, …
- Signal processing (peaks, spectra, …) and analysis simulation

Broad variety of challenges, e.g.

- High dimensionality and heterogeneity
- Sparse labelling
- Multiple descriptors, correlated features
- Many different data formats

Broad variety of tools!

Engel T. (2018) *Computer Processing of Chemical Structure Information* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 43-119), Wiley-VCH.

Useful tools: Open Babel

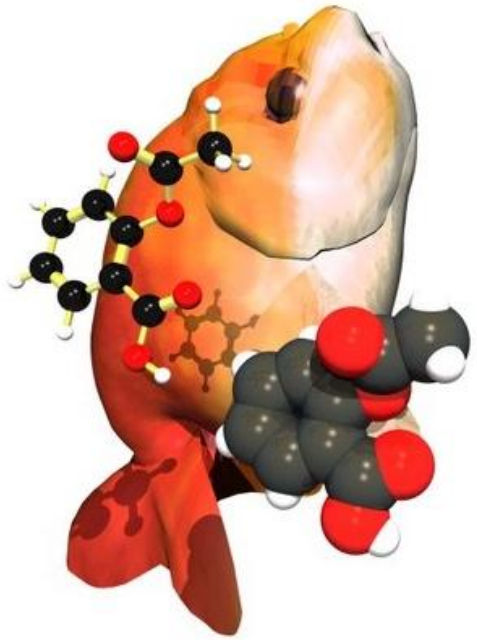
Chemical toolbox, mainly focused on structures:

- Ready-to-use programs, and complete programmer's toolkit
- Read, write and convert over 110 chemical file formats
- Filter and search molecular files using SMARTS and other methods

Many different file formats supported from various areas, e.g.

- Molecular modelling
- Chemistry
- Materials
- Biochemistry
- …

GUI and API



https://openbabel.org/

Useful tools: RDKit



Chemical cheminformatics toolbox:

- Molecular handling and representation
- **Substructure and similarity searches**
- **Molecular descriptors** & fingerprints
- Reaction encoding

Use case, e.g.

- Virtual screening and drug discovery
- QSAR/QSPR modeling (molecular properties → ML models)
- Molecular similarity and clustering
- Reaction prediction and cheminformatics pipelines
- Integration with deep learning frameworks (via DeepChem, PyTorch Geometric)

https://www.rdkit.org/

Useful tools: BioPandas



Python library extending pandas to handle .pdb files:

- Converts PDB and MOL2 structural data into pandas DataFrames - easy to filter, group, and analyse
- Convenient coordinate access
- Selection & filtering: Access subset atoms, residues, or chains
- Vectorised operations: Use pandas/numpy syntax for efficient numeric computations on molecular data
- Visualisation-ready: Plays well with matplotlib or seaborn

```
# Initialize a new PandasPdb object
# and fetch the PDB file from rcsb.org
>>> from biopandas.pdb import PandasPdb
>>> ppdb = PandasPdb().fetch_pdb('3eiy')
>>> ppdb.df['ATOM'].head()
```

	record_name	atom_number	...	atom_name	...	x_coord	y_coord	z_coord	...
0	ATOM	1		N		2.527	54.656	-1.667	
1	ATOM	2		CA		3.259	54.783	-0.368	
2	ATOM	3		C		4.127	53.553	-0.105	
3	ATOM	4		O		5.274	53.451	-0.594	
4	ATOM	5		CB		2.273	54.944	0.792	

Use cases:

- Compute distances, angles, or centroids between atoms or residues
- Analyze protein–ligand interfaces
- Extract subsets of atoms for simulations or feature calculation
- Combine structure data with bioinformatics or experimental data in panda

https://biopandas.github.io/biopandas/

Useful tools: BioPython



Python library for computational biology and bioinformatics:

- Sequence Handling: Work with DNA, RNA, and protein sequences using Seq objects
- Translate, transcribe, and manipulate biological sequences easily
- File Parsing: Read / write a wide range of file formats (FASTA, GenBank, PDB, …)
- Database Access: Retrieve bioinformatics data (e.g., NCBI Entrez, UniProt)
- Structural Biology: Parse and analyse PDB structures (though less DataFrame-oriented than BioPandas)

Use cases:

- Sequence analysis and annotation
- BLAST queries and parsing results
- Reading/writing genomic or proteomic data
- Working with alignments and evolutionary trees
- Integrating biological data in ML workflows.

```
>>>from Bio.Seq import Seq
>>>dna_sequence = Seq("AGGCTTCTCGTA")
>>>print(dna_sequence)
Seq('AGGCTTCTCGTA')

>>>print(dna_sequence[2:7])
Seq('GCTTC')

>>>print(dna_sequence.reverse_complement())
Seq('TACGAGAAGCCT')

>>>rna_sequence = dna_sequence.transcribe()
>>>print(rna_sequence)
Seq('AGGCUUCUCGUA')

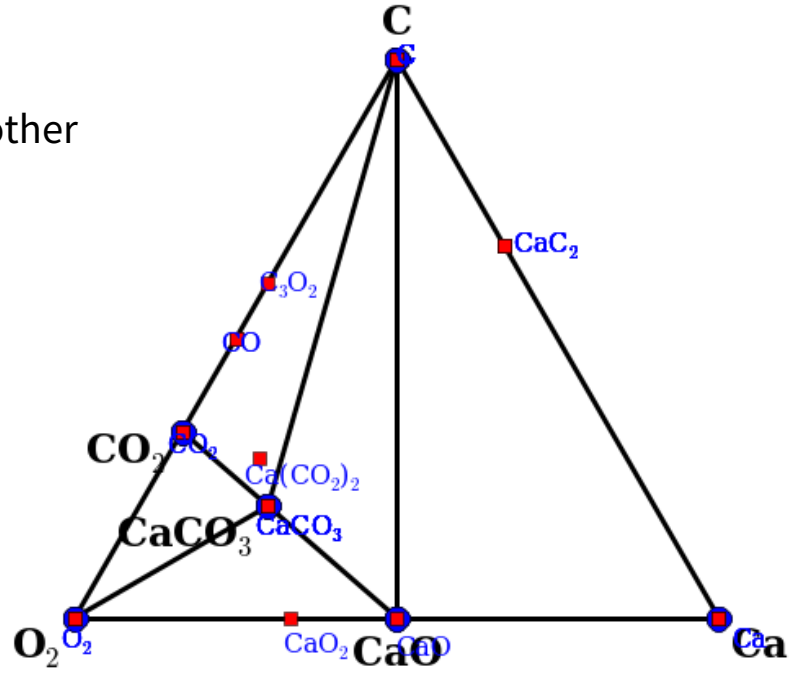
>>>print(rna_sequence.translate())
Seq('RLLV')
```

https://biopython.org/

Useful tools: PyMatGen

Python library for materials analysis (Materials Project!). Main features include:

- Flexible classes for representing Element, Site, Molecule and Structure objects
- File Parsing: Read and write a wide range of relevant file formats (VASP, ABINIT, CIF, Gaussian, XYZ, …)
- Generate and manipulate surfaces, apply symmetry operations and lattice transformations
- Electronic structure analyses, e.g. density of states
- Tools for plotting band structures and phase diagrams
- Integration with the Materials Project REST API, Crystallography Open Database and other external data sources.



Use cases:

- Pre- and post-processing for DFT calculations
- Materials informatics and data-driven discovery
- Building custom computational workflows
- Integrating with machine learning pipelines for materials prediction

https://pymatgen.org/

Activity: RDKit Workshop

1) Load and explore chemical datasets (Delaney solubility dataset)
2) Compute RDKit descriptors
3) Build regression model for molecular property prediction (data quality assessment)
4) Calculate fingerprints
5) Measure structural similarity (Tanimoto)
6) Cluster molecules by structure
7) Identify and query substructures using SMARTS patterns